



Hessian Lane-Emden Type Systems with Measures Involving Sub-natural Growth Terms

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Received: 17 November 2023 / Accepted: 8 September 2024 / Published online: 28 September 2024
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Abstract

We are interested in a class of k -Hessian Lane-Emden type systems with measure data in the “sublinear growth” rate. We give global pointwise estimates of the so-called Brezis-Kamin type in terms of Wolff potentials, which allows us to obtain necessary and sufficient conditions for the existence of positive solutions. The approach allows one to treat other nonlinear elliptic problems, such as equations involving general quasilinear operators and fractional Laplacian.

Keywords k -Hessian equations · Lane-Emden system · Integral equations · Wolff potentials · Measure data · Positive Solution

Mathematics Subject Classification (2010) 35J60 · 35B09 · 35B45 · 31C15 · 35R06 · 45G15

1 Introduction

The problem of existence for a class of fully nonlinear elliptic systems of equations of the Lane-Emden type has been addressed. Furthermore, comprehensive pointwise estimates of solutions have been attained. This achievement encompasses the resolution of the following class of problems:

$$\begin{cases} F_k[-u] = \sigma v^{q_1}, & u > 0 \text{ in } \mathbb{R}^n, \\ F_k[-v] = \sigma u^{q_2}, & v > 0 \text{ in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} u(x) = 0, & \lim_{|x| \rightarrow \infty} v(x) = 0, \end{cases} \quad (S)$$

where $n \geq 3$ and σ belongs to $M^+(\mathbb{R}^n)$, the set of all nonnegative Radon measures on $\mathbb{R}^n$. Here, $F_k[u]$ is the k -Hessian operator for $1 \leq k < n/2$ defined as the sum of $k \times k$ minors of the Hessian matrix D^2u . We analyze Eq. [S](#) for the sub-natural growth, that is, $0 < q_i < k$

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for $i = 1, 2$. We can see that when $u = v$ and $q_1 = q_2 = q$, Syst. Eq. [S](#) can be reduced to the following class of problems involving a single equation

$$\begin{cases} F_k[-u] = \sigma u^q & \text{in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} u(x) = 0. \end{cases} \quad (P)$$

Sublinear elliptic equations with measurable coefficients have been investigated by many authors. In [\[2\]](#), H. Brezis and S. Kamin established a necessary and sufficient condition for the existence of a bounded solution to the sublinear problem

$$-\Delta u = \sigma(x)u^q \quad \text{in } \mathbb{R}^n \quad (n \geq 3) \quad (1.1)$$

with $0 < q < 1$, $0 \not\equiv \sigma \geq 0$ and $\sigma \in L_{\text{loc}}^\infty(\mathbb{R})$. For bounded solutions of Eq. [1.1](#), by using the Green function of the Laplacian in balls of $\mathbb{R}^n$, they proved global pointwise estimates of the form

$$c^{-1}(\mathbf{I}_2\sigma)^{\frac{1}{1-q}} \leq u \leq c\mathbf{I}_2\sigma, \quad (1.2)$$

where $c > 0$ is a constant independent of u and

$$\mathbf{I}_2\sigma(x) = \int_{\mathbb{R}^n} \frac{d\sigma(y)}{|x-y|^{n-2}}$$

is the Newtonian potential of σ . See also [\[3, 4\]](#) and the references given there for the existence of a larger class of solutions satisfying Brezis-Kamin type estimates in terms of Wolff potentials to nonlinear elliptic problems involving p -Laplace and fractional Laplace operators.

There are some challenges in studying Hessian Lane-Emden type systems with measures of the form Eq. [S](#) because our criteria for the solvability of Hessian equations on the entire space depends on the existence result in subdomains with specific geometric conditions. Furthermore, we need to develop accurate estimates that are more involved than the ones we can find in the literature. To overcome these difficulties, instead of the use of Green functions, we used a method based on some elements of nonlinear potential theory, accurately on the systematic use of Wolff potentials

$$\mathbf{W}_{\alpha,p}\sigma(x) = \int_0^\infty \left(\frac{\sigma(B(x,t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t}, \quad (1.3)$$

where $\sigma \in M^+(\mathbb{R}^n)$, $1 < p < \infty$, $0 < \alpha < n/p$ and $B(x, t)$ is the open ball of radius $t > 0$ centered at $x \in \mathbb{R}^n$. Observe that $\mathbf{W}_{\alpha,p}\sigma$ is always positive for $\sigma \not\equiv 0$ which may be ∞ . For more details, we refer the reader to [\[5, 7–9, 11\]](#) and references therein.

To examine fully nonlinear elliptic systems of the form Eq. [S](#), first, in a general framework, we study the existence of $(v, u) \in L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma)$ satisfying pointwise the following system of integral equations

$$\begin{cases} u = \mathbf{W}_{\alpha,p}(v^{q_1}d\sigma), & d\sigma\text{-a.e in } \mathbb{R}^n, \\ v = \mathbf{W}_{\alpha,p}(u^{q_2}d\sigma), & d\sigma\text{-a.e in } \mathbb{R}^n, \end{cases} \quad (SI)$$

where u and v constitute a pair of nonnegative functions. We prove the existence of solutions to Syst. Eq. [S](#) by using the method of successive approximations. We use a solution of Eq. [SI](#) as an upper bound for Syst. Eq. [S](#). Throughout this work, solutions to Eq. [S](#) will be treated as k -subharmonic (k -convex) functions (for details, see Sect. [3](#)).

Wolff potentials with $\alpha = 2k/(k+1)$ and $p = k+1$, for $k < n/2$, in the context of fully nonlinear problems, was considered in [10], where D. A. Labutin proved that if $\mu \in M^+(\mathbb{R}^n)$ and w is a nonnegative k -subharmonic solution of the equation

$$\begin{cases} F_k[-w] = \mu & \text{in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} w(x) = 0, \end{cases}$$

then there exists a constant $K = K(n, k) \geq 1$ such that

$$K^{-1} \mathbf{W}_{\frac{2k}{k+1}, k+1} \mu(x) \leq w(x) \leq K \mathbf{W}_{\frac{2k}{k+1}, k+1} \mu(x) \quad \forall x \in \mathbb{R}^n.$$

In particular, such w exists if and only if $\mathbf{W}_{2k/(k-1), k+1} \mu < \infty$, or equivalently

$$\int_1^\infty \left(\frac{\mu(B(0, t))}{t^{n-2k}} \right)^{\frac{1}{k}} \frac{dt}{t} < \infty. \quad (1.4)$$

For the next two theorems, we assume that $\sigma \in M^+(\mathbb{R}^n)$ satisfies $\mathbf{W}_{\alpha, p} \sigma(x) < \infty$ for almost every $x \in \mathbb{R}^n$, or equivalently by [3, Corollary 3.2],

$$\int_1^\infty \left(\frac{\sigma(B(0, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} < \infty. \quad (1.5)$$

This means that $\mathbf{W}_{\alpha, p} \sigma(x) < \infty$ for almost every $x \in \mathbb{R}^n$ if and only if $\mathbf{W}_{\alpha, p} \sigma(x_0) < \infty$ for some $x_0 \in \mathbb{R}^n$.

1.1 Main Results

From now on, we assume that $1 < p < \infty$, $0 < q_i < p-1$, $i = 1, 2$, $0 < \alpha < n/p$ and $\sigma \in M^+(\mathbb{R}^n)$. To state our first result in a precise way, let us recall the notion of Riesz capacity of a compact set $E \subset \mathbb{R}^n$,

$$\text{cap}_{\mathbf{I}_\alpha, p}(E) = \inf \{ \|f\|_{L^p}^p : f \in L^p(\mathbb{R}^n), f \geq 0, \mathbf{I}_\alpha f \geq 1 \text{ on } E \}, \quad (1.6)$$

where $\mathbf{I}_\alpha \sigma$ is the Riesz potential of order α defined for $0 < \alpha < n$ by

$$\mathbf{I}_\alpha \sigma(x) = \int_{\mathbb{R}^n} \frac{d\sigma(y)}{|x-y|^{n-\alpha}}, \quad x \in \mathbb{R}^n. \quad (1.7)$$

The next condition on σ is the key to construct solutions to Eq. [SI](#). There exists $C_\sigma > 0$ such that

$$\sigma(E) \leq C_\sigma \text{cap}_{\mathbf{I}_\alpha, p}(E) \quad \text{for all compact sets } E \subset \mathbb{R}^n. \quad (1.8)$$

Condition Eq. [1.8](#) is equivalent to: for any $r > 0$, there exists $c = c_\sigma > 0$, for which it holds

$$\int_E (\mathbf{W}_{\alpha, p} \sigma_E)^r d\sigma \leq c_\sigma \sigma(E) \quad \text{for all compact sets } E \subset \mathbb{R}^n, \quad (1.9)$$

where $\sigma_E = \chi_E \sigma$ (see [4, Lemma 2.1] for more details). It is through this equivalence that we will be able to exhibit a nontrivial solution to Eq. [SI](#).

Theorem 1.1 *Assuem $\sigma \in M^+(\mathbb{R}^n)$ satisfies Eqs. [1.5](#) and [1.8](#). Then, there exists a solution (u, v) to Syst. Eq. [SI](#) such that*

$$c^{-1} (\mathbf{W}_{\alpha, p} \sigma)^{\frac{(p-1)(p-1+q_1)}{(p-1)^2-q_1q_2}} \leq u \leq c \left(\mathbf{W}_{\alpha, p} \sigma + (\mathbf{W}_{\alpha, p} \sigma)^{\frac{(p-1)(p-1+q_1)}{(p-1)^2-q_1q_2}} \right),$$

$$c^{-1} (\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_2)}{(p-1)^2-q_1q_2}} \leq v \leq c \left(\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_2)}{(p-1)^2-q_1q_2}} \right), \quad (1.10)$$

where $c = c(n, p, q_1, q_2, \alpha, C_\sigma) > 0$. In addition, $u, v \in L^r_{\text{loc}}(\mathbb{R}^n, d\sigma)$ for all $r > 0$.

Remark 1.2 We use assumption Eq. 1.8 in the proof of Eq. 1.10 to conclude the upper bounds estimate, and this assumption is not used to conclude the lower bounds estimate.

The next theorem brings a necessary and sufficient on σ to the existence of solutions (u, v) to Eq. SI which enjoy the bilateral estimates Eq. 1.10. Suppose $\sigma \in M^+(\mathbb{R}^n)$ satisfies

$$\begin{aligned} \mathbf{W}_{\alpha,p} \left((\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_1)q_2}{(p-1)^2-q_1q_2}} d\sigma \right) &\leq \lambda \left(\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_1)}{(p-1)^2-q_1q_2}} \right), \\ \mathbf{W}_{\alpha,p} \left((\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_2)q_1}{(p-1)^2-q_1q_2}} d\sigma \right) &\leq \lambda \left(\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\frac{(p-1)(p-1+q_2)}{(p-1)^2-q_1q_2}} \right), \end{aligned} \quad (1.11)$$

here the right-hand sides are finite almost everywhere in $\mathbb{R}^n$, and λ is a positive constant. In general, condition Eq. 1.11 is weaker than Eq. 1.8 (see, for instance, [4]).

Theorem 1.3 Suppose $\sigma \in M^+(\mathbb{R}^n)$ satisfying Eq. 1.11 for some $\lambda > 0$. Then Syst. Eq. SI possesses a nontrivial solution (u, v) satisfying the bilateral estimates Eq. 1.10, where $c = c(n, p, \alpha, q_1, q_2, \lambda) > 0$. Conversely, if (u, v) is a solution to Eq. SI satisfying Eq. 1.10 for some $c > 0$, then Eq. 1.11 holds, where $\lambda = \lambda(p, q_1, q_2, \alpha, c) > 0$.

Remark 1.4 For the case $q_1 = q_2$, one can see that the solution (u, v) obtained in Theorem 1.1 is such that $u = v$, and it satisfies

$$c^{-1} (\mathbf{W}_{\alpha,p}\sigma)^{\frac{p-1}{p-1-q}} \leq u \leq c \left(\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\frac{p-1}{p-1-q}} \right),$$

where $q := q_1$ (see details in Sect. 2). Therefore [4, Theorem 3.2] is a corollary of Theorem 1.1 for $q_1 = q_2$. Similarly, [4, Theorem 3.3] is a corollary of Theorem 1.3, since in the case $q_1 = q_2$ condition Eq. 1.11 is written as

$$\mathbf{W}_{\alpha,p} \left((\mathbf{W}_{\alpha,p}\sigma)^{\frac{q(p-1)}{p-1-q}} d\sigma \right) \leq \lambda \left(\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\frac{p-1}{p-1-q}} \right) < \infty.$$

Next, we state an application of the above results to Syst. Eq. S. For now on, $\mathbf{W}_k$ will denote $\mathbf{W}_{2k/(k+1), k+1}$.

Theorem 1.5 Let $\sigma \in M^+(\mathbb{R}^n)$ satisfying Eqs. 1.5 and 1.8 with $\alpha = 2k/(k+1)$ and $p = k+1$. Then there exists a solution (u, v) to Syst. Eq. S satisfying

$$\begin{aligned} c^{-1} (\mathbf{W}_k\sigma)^{\frac{k(k+q_1)}{k^2-q_1q_2}} \leq u \leq c \left(\mathbf{W}_k\sigma + (\mathbf{W}_k\sigma)^{\frac{k(k+q_1)}{k^2-q_1q_2}} \right), \\ c^{-1} (\mathbf{W}_k\sigma)^{\frac{k(k+q_2)}{k^2-q_1q_2}} \leq v \leq c \left(\mathbf{W}_k\sigma + (\mathbf{W}_k\sigma)^{\frac{k(k+q_2)}{k^2-q_1q_2}} \right), \end{aligned} \quad (1.12)$$

where $c = c(n, k, q_1, q_2, \alpha, C_\sigma) > 0$. In addition, $u, v \in L^r_{\text{loc}}(\mathbb{R}^n, d\sigma)$, for all $r > 0$.

As in Theorem 1.3, we may assume a weaker condition on σ than Eq. 1.8, with $\alpha = 2k/(k+1)$ and $p = k+1$, and nonetheless assure the existence of solutions to Syst. Eq. S such that Eq. 1.12 holds. The following theorem deals with this.

Theorem 1.6 Let $1 \leq k < n/2$, $0 < q_i < k$, $i = 1, 2$ and $\sigma \in M^+(\mathbb{R}^n)$.

(i) If there exists a solution (u, v) to Syst. Eq. [S](#) satisfying Eq. [1.12](#) for some $c > 0$, then there exists a constant $\lambda = \lambda(n, k, q_1, q_2, c) > 0$ such that, for almost everywhere, it holds

$$\begin{aligned} \mathbf{W}_k \left((\mathbf{W}_k \sigma)^{\frac{k(k+q_1)q_2}{k^2-q_1q_2}} d\sigma \right) &\leq \lambda \left(\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\frac{k(k+q_1)}{k^2-q_1q_2}} \right) < \infty, \\ \mathbf{W}_k \left((\mathbf{W}_k \sigma)^{\frac{k(k+q_2)q_1}{k^2-q_1q_2}} d\sigma \right) &\leq \lambda \left(\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\frac{k(k+q_2)}{k^2-q_1q_2}} \right) < \infty. \end{aligned} \quad (1.13)$$

(ii) Suppose that σ satisfies Eq. [1.13](#) for some $\lambda > 0$. Then there exists a solution (u, v) to Syst. Eq. [S](#) satisfying Eq. [1.12](#), where $c = c(n, k, q_1, q_2, \lambda)$.

From the result above, we conclude that condition Eq. [1.13](#) is necessary and sufficient for the existence of a solution to Syst. Eq. [S](#) satisfying Eq. [1.12](#).

The following result is a simple consequence of the preceding theorems by taking $q_1 = q_2$. We highlight that it was already announced in [[3](#), [4](#)].

Corollary 1.7 Let $1 \leq k < n/2$, $0 < q < k$ and $\sigma \in M^+(\mathbb{R}^n)$.

(i) If Eqs. [1.4](#) and [1.8](#) hold with $\alpha = 2k/(k+1)$ and $p = k+1$, then there exists a k -subharmonic nonnegative solution to Eq. [P](#) satisfying

$$c^{-1} (\mathbf{W}_k \sigma)^{\frac{k}{k-q}} \leq u \leq c \left(\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\frac{k}{k-q}} \right), \quad (1.14)$$

where $c = c(n, k, q, \sigma) > 0$.

(ii) Let u be a k -subharmonic nonnegative solution to Eq. [P](#), satisfying Eq. [1.14](#). Then there exists a constant $\lambda = \lambda(n, k, q, c) > 0$ such that, for almost everywhere, it holds

$$\mathbf{W}_k \left((\mathbf{W}_k \sigma)^{\frac{kq}{k-q}} d\sigma \right) \leq \lambda \left(\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\frac{k}{k-q}} \right) < \infty. \quad (1.15)$$

(iii) Suppose that σ satisfies Eq. [1.15](#) for some $\lambda > 0$. Then there exists a k -subharmonic nonnegative solution to Eq. [P](#) satisfying Eq. [1.14](#), where $c = c(n, k, q, \lambda) > 0$.

Let us mention that Corollary [1.7](#) is an extension of [[2](#), Theorem 1]. Indeed, let $\sigma \in M^+(\mathbb{R}^n)$ such that $\mathbf{I}_2 \sigma \in L^\infty(\mathbb{R}^n)$, as in [[2](#)]. Since $\mathbf{I}_2 \sigma = (n-2)\mathbf{W}_1 \sigma$, one has for all compact set E

$$\sigma(E) = \int_E d\sigma = \int_E \mathbf{I}_2 \sigma \frac{d\sigma}{\mathbf{I}_2 \sigma} \leq \frac{\|\mathbf{I}_2 \sigma\|_{L^\infty}}{n-2} \int_E \frac{d\sigma}{\mathbf{W}_1 \sigma}.$$

By [[18](#), Theorem 1.11] there exists $C = C(n) > 0$ such that

$$\int_E \frac{d\sigma}{\mathbf{W}_1 \sigma} \leq C \operatorname{cap}_{\mathbf{I}_1, 2}(E).$$

Hence $\sigma(E) \leq \tilde{C} \|\mathbf{I}_2 \sigma\|_{L^\infty} \operatorname{cap}_{\mathbf{I}_1, 2}(E)$, and condition Eq. [1.8](#) follows for $\alpha = 1$ and $p = 2$. Thus there exists a bounded solution $u \geq 0$ to Eq. [P](#) with $k = 1$, that is

$$-\Delta u = \sigma u^q \quad \text{in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} u(x) = 0,$$

satisfying the so-called Brezis-Kamin estimates Eq. [1.2](#), or more precise

$$c^{-1} (\mathbf{I}_2 \sigma)^{\frac{1}{1-q}} \leq u \leq c \left(\mathbf{I}_2 \sigma + (\mathbf{I}_2 \sigma)^{\frac{1}{1-q}} \right),$$

whenever $\mathbf{I}_2\sigma$ is bounded.

Recently, studies of systems involving the k -Hessian operator have been done. See for example [1, 19, 20] and references therein. In measure setting, M. Véron, Q. Nguyen and L. Véron, in [1], proposed to study nonlinear systems with super-natural growth in $\mathbb{R}^n$, as a counterpart of [12]. They could give necessary and sufficient conditions for existence expressed in terms of Riesz and Bessel capacities, together with bilateral estimates in terms of Wolff potentials. We mention that in [6], we applied the main argument of this work to study systems of quasilinear equations with sub-natural growth in $\mathbb{R}^n$.

1.2 Organization of the Paper

In Sect. 2, we produce a framework to solve Syst. Eq. [5I](#) by proving Theorem [1.1](#) and Theorem [1.3](#). In Sect. 3, we present auxiliary results and definitions from k -subharmonic functions (or k -convex functions). In Sect. 4, we provide the proofs of Theorems [1.5](#), Theorem [1.6](#), and Corollary [1.7](#).

1.3 Notation

We use the following notation. Let Ω be a domain in $\mathbb{R}^n$.

- $\chi_E :=$ the characteristic function of a set E .
- We use the Greek letters μ and ω to denote Radon measures.
- $C_c^\infty(\Omega) :=$ the set of all infinitely differentiable functions with compact support in Ω .
- $C(\Omega) :=$ the set of all continuous functions in on Ω .
- $L_{\text{loc}}^s(\Omega, d\mu) :=$ the local L^s space with respect to $\mu \in M^+(\Omega)$, $s > 0$. If μ is the Lebesgue measure, we write $L_{\text{loc}}^s(\Omega)$.
- We denote by $\sigma(E) = \int_E d\sigma$ the measure of any σ -measurable subset E of Ω .
- The letters $c, \tilde{c}, C$, and $\tilde{C}$ denote various constants with and without indices which may change from line to line.

2 Systems of Wolff Potentials

To prove Theorem [1.1](#) and Theorem [1.3](#), we will use the following technical lemma, which is a direct consequence of Eq. [1.9](#). First, notice that condition Eq. [1.9](#) assure that, for all $r > 0$,

$$\int_B (\mathbf{W}_{\alpha,p}\sigma_B)^r d\sigma \leq c_\sigma \sigma(B), \quad \text{for all balls } B \subset \mathbb{R}^n. \quad (2.1)$$

Lemma 2.1 *Let $\sigma \in M^+(\mathbb{R}^n)$ satisfying Eqs. [1.5](#) and [1.9](#). Then, for all $r > 0$, it holds*

$$\mathbf{W}_{\alpha,p}\sigma \in L_{\text{loc}}^r(\mathbb{R}^n, d\sigma).$$

Proof It is enough to prove that $\mathbf{W}_{\alpha,p}\sigma \in L^r(B, d\sigma)$, for all balls $B \subset \mathbb{R}^n$. Fixed $B = B(x_0, R)$, setting $2B = B(x_0, 2R)$ and denoting by $(2B)^c$ the complement of $2B$ in $\mathbb{R}^n$, we can write

$$\int_B (\mathbf{W}_{\alpha,p}\sigma)^r d\sigma = \int_B \left(\int_0^\infty \left(\frac{(\sigma_{2B} + \sigma_{(2B)^c})(B(x, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} dt \right)^r d\sigma$$

$$= \int_B \left(\int_0^\infty \left(\frac{\sigma_{2B}(B(x, t)) + \sigma_{(2B)^c}(B(x, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} dt \right)^r d\sigma.$$

Thus,

$$\begin{aligned} \int_B (\mathbf{W}_{\alpha, p\sigma})^r d\sigma &\leq 2^{\frac{r}{p-1}} \int_B \left(\int_0^\infty \left(\frac{\sigma_{2B}(B(x, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} dt + \int_0^\infty \left(\frac{\sigma_{(2B)^c}(B(x, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} dt \right)^r d\sigma \\ &\leq c_1(I_1 + I_2), \end{aligned}$$

where

$$I_1 := \int_B (\mathbf{W}_{\alpha, p\sigma_{2B}})^r d\sigma \quad \text{and} \quad I_2 := \int_B (\mathbf{W}_{\alpha, p\sigma_{(2B)^c}})^r d\sigma$$

and $c_1 = c_1(p, r) > 0$. The estimate of I_1 is a consequence of Eq. 2.1. Indeed,

$$I_1 \leq c_1 \int_{2B} (\mathbf{W}_{\alpha, p\sigma_{2B}})^r d\sigma \leq c_2 \sigma(2B) < \infty,$$

where $c_2 = c_2(n, p, r, \alpha, c_\sigma)$.

To estimate I_2 , we first observe that $(2B)^c \cap B(y, t) = \emptyset$ for $y \in B$ and $0 < t < R$, whence for $y \in B$

$$\begin{aligned} \mathbf{W}_{\alpha, p\sigma_{(2B)^c}}(y) &= \int_0^\infty \left(\frac{\sigma((2B)^c \cap B(y, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \\ &= \int_R^\infty \left(\frac{\sigma((2B)^c \cap B(y, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \leq \int_R^\infty \left(\frac{\sigma(B(y, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t}. \end{aligned}$$

Since $B(y, t) \subset B(0, 2t)$ for $y \in B$ and $t \geq R$, it follows

$$\mathbf{W}_{\alpha, p\sigma_{(2B)^c}}(y) \leq \int_R^\infty \left(\frac{\sigma(B(0, 2t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \leq 2^{\frac{n-\alpha p}{p-1}} \int_R^\infty \left(\frac{\sigma(B(0, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t}.$$

Applying the previous inequality,

$$I_2 = c_1 \int_B (\mathbf{W}_{\alpha, p\sigma_{(2B)^c}})^r d\sigma \leq c_3 \left(\int_R^\infty \left(\frac{\sigma(B(0, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \right)^r \sigma(B) < \infty.$$

This completes the proof of Lemma 2.1. $\square$

Before proving Theorem 1.1, we recall the following result [3, Lemma 3.5].

Lemma A *Let $\omega \in M^+(\mathbb{R}^n)$. Then, there exists a constant $\kappa = \kappa(n, p, \alpha) > 0$ such that, for all $s > 0$ and for all $x \in \mathbb{R}^n$, it holds*

$$\mathbf{W}_{\alpha, p}((\mathbf{W}_{\alpha, p}\omega)^s d\omega)(x) \geq \kappa^{\frac{s}{p-1}} (\mathbf{W}_{\alpha, p}\omega(x))^{\frac{s}{p-1}+1}.$$

2.1 Proof of Theorem 1.1

The proof runs by the sub-supersolution method. The notion of *supersolution* or *subsolution* to Eq. 1.1 will be used, where they are defined by replacing “=” by “ $\geq$ ” or “ $\leq$ ” in Eq. 1.1,

respectively. First, we set

$$\gamma_1 = \frac{(p-1)(p-1+q_1)}{(p-1)^2 - q_1 q_2} \quad \text{and} \quad \gamma_2 = \frac{(p-1)(p-1+q_2)}{(p-1)^2 - q_1 q_2}, \quad (2.2)$$

which can be rewritten as follows

$$\gamma_1 = \frac{q_1}{p-1} \gamma_2 + 1 \quad \text{and} \quad \gamma_2 = \frac{q_2}{p-1} \gamma_1 + 1. \quad (2.3)$$

Claim 1 There exists a constant $\lambda_1 > 0$ sufficiently small such that setting

$$\underline{u} = \lambda_1 (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1} \quad \text{and} \quad \underline{v} = \lambda_1 (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2},$$

the pair $(\underline{u}, \underline{v})$ is a subsolution to Eq. [SI](#).

Indeed, by Lemma [A](#) and Eq. [2.3](#), it follows

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\underline{v}^{q_1} d\sigma) &= \lambda_1^{\frac{q_1}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{q_1 \gamma_2} d\sigma) \geq \lambda_1^{\frac{q_1}{p-1}} \kappa^{\frac{q_1}{p-1} \gamma_2} (\mathbf{W}_{\alpha,p} \sigma)^{\frac{q_1}{p-1} \gamma_2 + 1} \\ &= \lambda_1^{\frac{q_1}{p-1}} \kappa^{\frac{q_1}{p-1} \gamma_2} (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1}, \\ \mathbf{W}_{\alpha,p}(\underline{u}^{q_2} d\sigma) &= \lambda_1^{\frac{q_2}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{q_2 \gamma_1} d\sigma) \geq \lambda_1^{\frac{q_2}{p-1}} \kappa^{\frac{q_2}{p-1} \gamma_1} (\mathbf{W}_{\alpha,p} \sigma)^{\frac{q_2}{p-1} \gamma_1 + 1} \\ &= \lambda_1^{\frac{q_2}{p-1}} \kappa^{\frac{q_2}{p-1} \gamma_1} (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2}. \end{aligned}$$

Thus, taking $\lambda_1 = \min\{1, \kappa^{(q_1 \gamma_2)/(p-1-q_1)}, \kappa^{(q_2 \gamma_1)/(p-1-q_2)}\}$, we concluded that

$$\underline{u} \leq \mathbf{W}_{\alpha,p}(\underline{v}^{q_1} d\sigma) \quad \text{and} \quad \underline{v} \leq \mathbf{W}_{\alpha,p}(\underline{u}^{q_2} d\sigma),$$

which proves Claim 1.

To obtain a supersolution to Eq. [SI](#) we define

$$\bar{u} = \lambda_2 (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1}) \quad \text{and} \quad \bar{v} = \lambda_2 (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2}).$$

We are going to prove that taking $\lambda_2 > 0$ sufficiently large we have that $(\bar{u}, \bar{v})$ is a supersolution to Eq. [SI](#), such that $\underline{u} \leq \bar{u}$ and $\underline{v} \leq \bar{v}$. This proof will be divided into two steps. First, we estimate $\mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma)$,

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma)(x) &\leq \\ \lambda_2^{\frac{q_1}{p-1}} c_1 &\left(\int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p} \sigma)^{q_1} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} + \int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p} \sigma)^{q_1 \gamma_2} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \right), \end{aligned} \quad (2.4)$$

for some $c_1 = c_1(p, q_1) > 0$. Analysis similar to that in the proof of Lemma [2.1](#) shows that

$$\begin{aligned} \int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p} \sigma)^{q_1} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} &\leq \int_0^\infty \left(\frac{c_2 \sigma(B(x, 2t)) + c_2 \sigma(B(x, t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \\ &\leq c_3 \int_0^\infty \left(\frac{\sigma(B(x, 2t))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} + c_3 \mathbf{W}_{\alpha,p} \sigma(x) \\ &\leq c_4 \mathbf{W}_{\alpha,p} \sigma(x), \end{aligned} \quad (2.5)$$

where c_2 , c_3 and c_4 depend only on n , p , q_1 and σ . Next, we write

$$\begin{aligned} \int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma &\leq c_5 \int_{B(x,t)} \left[\int_0^t \left(\frac{\sigma(B(y,s))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} d\sigma(y) \\ &\quad + c_5 \int_{B(x,t)} \left[\int_t^\infty \left(\frac{\sigma(B(y,s))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} d\sigma(y) \\ &=: c_5 (I_1 + I_2), \end{aligned}$$

where $c_5 = c_5(p, q_1, q_2) > 0$. Since $B(y, s) \subset B(x, 2t)$ for $y \in B(x, t)$ and $s \leq t$, it follows from Eq. 2.1 that

$$\begin{aligned} I_1 &\leq \int_{B(x,2t)} \left[\int_0^t \left(\frac{\sigma(B(y,s) \cap B(x,2t))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} d\sigma(y) \\ &\leq \int_{B(x,2t)} [\mathbf{W}_{\alpha,p}\sigma_{B(x,2t)}]^{\gamma_2 q_1} d\sigma(y) \leq c_6 \sigma(B(x, 2t)), \end{aligned}$$

where $c_6 = c_6(n, p, q_1, q_2, C_\sigma)$. If $s \geq t$ and $y \in B(x, t)$, then $B(y, s) \subset B(x, 2s)$, whence

$$\begin{aligned} I_2 &\leq \int_{B(x,t)} \left[\int_t^\infty \left(\frac{\sigma(B(x,2s))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} d\sigma(y) \\ &= \sigma(B(x, t)) \left[\int_t^\infty \left(\frac{\sigma(B(x,2s))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} \\ &\leq \sigma(B(x, t)) \left[\int_0^\infty \left(\frac{\sigma(B(x,2s))}{s^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{ds}{s} \right]^{\gamma_2 q_1} \leq c_7 \sigma(B(x, t)) (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_2 q_1}, \end{aligned}$$

where $c_7 = 2^{((n-\alpha p)\gamma_2 q_1)/(p-1)}$. This gives

$$\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma \leq c_8 [\sigma(B(x, 2t)) + \sigma(B(x, t)) (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_2 q_1}].$$

Applying the previous inequality and Eq. 2.5 in Eq. 2.4,

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma)(x) &\leq \lambda_2^{\frac{q_1}{p-1}} c_1 \left[c_4 \mathbf{W}_{\alpha,p}\sigma(x) \right. \\ &\quad \left. + \int_0^\infty \left(\frac{c_8 [\sigma(B(x, 2t)) + \sigma(B(x, t)) (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_2 q_1}]}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \right] \\ &\leq \lambda_2^{\frac{q_1}{p-1}} c_1 \left[c_4 \mathbf{W}_{\alpha,p}\sigma(x) + c_9 \mathbf{W}_{\alpha,p}\sigma(x) + c_9 (\mathbf{W}_{\alpha,p}\sigma(x))^{\frac{\gamma_2 q_1}{p-1} + 1} \right] \\ &\leq \lambda_2^{\frac{q_1}{p-1}} c_{10} \left[\mathbf{W}_{\alpha,p}\sigma(x) + (\mathbf{W}_{\alpha,p}\sigma(x))^{\frac{\gamma_2 q_1}{p-1} + 1} \right]. \end{aligned} \quad (2.6)$$

We now apply this argument again to estimate $\mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma)$. Replacing q_1 by q_2 and γ_2 by γ_1 , we obtain

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma)(x) &\leq \\ \lambda_2^{\frac{q_2}{p-1}} \tilde{c}_1 &\left(\int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_2} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} + \int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_2 \gamma_1} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \right), \end{aligned} \quad (2.7)$$

with

$$\int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_2} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \leq \tilde{c}_2 \mathbf{W}_{\alpha,p}\sigma(x),$$

$$\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_2\gamma_1} d\sigma \leq \tilde{c}_3 [\sigma(B(x, 2t)) + \sigma(B(x, t))(\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_1 q_2}],$$

where $\tilde{c}_1$, $\tilde{c}_2$ and $\tilde{c}_3$ depend only on n , p , q_1 , q_2 and C_σ , given in Eq. 1.8. Hence

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma)(x) &\leq \lambda_2^{\frac{q_2}{p-1}} \tilde{c}_1 \left[\tilde{c}_2 \mathbf{W}_{\alpha,p}\sigma(x) \right. \\ &\quad \left. + \int_0^\infty \left(\frac{\tilde{c}_3 [\sigma(B(x, 2t)) + \sigma(B(x, t))(\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_1 q_2}]}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \right] \\ &\leq \lambda_2^{\frac{q_2}{p-1}} \tilde{c}_1 \left[\tilde{c}_2 \mathbf{W}_{\alpha,p}\sigma(x) + \tilde{c}_4 \mathbf{W}_{\alpha,p}\sigma(x) + \tilde{c}_4 (\mathbf{W}_{\alpha,p}\sigma(x))^{\frac{\gamma_1 q_2}{p-1} + 1} \right] \\ &\leq \lambda_2^{\frac{q_1}{p-1}} \tilde{c}_5 \left[\mathbf{W}_{\alpha,p}\sigma(x) + (\mathbf{W}_{\alpha,p}\sigma(x))^{\frac{\gamma_2 q_1}{p-1} + 1} \right]. \end{aligned} \quad (2.8)$$

Thus, taking λ_2 such that

$$\lambda_2 = \max\{\lambda_1, c_{10}^{\frac{p-1}{p-1-q_1}}, (\tilde{c}_5)^{\frac{p-1}{p-1-q_2}}\},$$

we deduce from Eqs. 2.3, 2.6 and 2.8 that

$$\begin{aligned} \bar{u} &\geq \underline{u}, & \bar{u} &\geq \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma), \\ \bar{v} &\geq \underline{v}, & \bar{v} &\geq \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma), \end{aligned} \quad (2.9)$$

which is our assertion.

We now construct a sequence of iterations (u_j, v_j) , for $j = 0, 1, 2, \dots$, where u_j and v_j are real functions on $\mathbb{R}^n$, given by

$$\begin{cases} (u_0, v_0) = (\underline{u}, \underline{v}) = (\lambda_1 (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}, \lambda_1 (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2}), \\ (u_j, v_j) = (\mathbf{W}_{\alpha,p}(u_{j-1}^{q_1} d\sigma), \mathbf{W}_{\alpha,p}(u_{j-1}^{q_1} d\sigma)), & j = 1, 2, \dots \end{cases} \quad (2.10)$$

Using Claim 1, $u_1 \geq u_0$ and $v_1 \geq v_0$. In addition, $u_1 \leq \bar{u}$ and $v_1 \leq \bar{v}$ by Eq. 2.9. Consequently, by induction, we can assert that the sequences $\{u_j\}$ and $\{v_j\}$ are nondecreasing (pointwise), with $\underline{u} \leq u_j \leq \bar{u}$ and $\underline{v} \leq v_j \leq \bar{v}$, for $j = 0, 1, \dots$. In view of Lemma 2.1, we have $(u_j, v_j) \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$ for all $j = 0, 1, \dots$, since $(\underline{u}, \underline{v}), (\bar{u}, \bar{v}) \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$.

Letting $j \rightarrow \infty$ in Eq. 2.10, from the Monotone Convergence Theorem, we deduce that there exist nonnegative functions $u = \lim_{j \rightarrow \infty} u_j$ and $v = \lim_{j \rightarrow \infty} v_j$ (pointwise) such that $(u, v) \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$, where the pair (u, v) satisfies Eq. SI with $\underline{u} \leq u \leq \bar{u}$ and $\underline{v} \leq v \leq \bar{v}$. Thus there exists $c = c(n, p, \alpha, q_1, q_2, C_\sigma) > 0$ such that (u, v) satisfies Eq. 1.10. This proves Theorem 1.1.

2.2 Proof of Theorem 1.3

Suppose that Eq. 1.11 holds for some $\lambda > 0$. As in Claim 1, let $\underline{u} = \lambda_1 (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}$ and $\underline{v} = \lambda_1 (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2}$, where γ_1 and γ_2 are given in Eq. 2.2. It was already proved that $(\underline{u}, \underline{v})$ is a subsolution to Eq. SI if $\lambda_1 > 0$ is chosen to be sufficiently small. We only need to establish a supersolution to Eq. SI under condition Eq. 1.11.

We claim that there exists $\lambda_2 > 0$ sufficiently large such that

$$(\bar{u}, \bar{v}) = (\lambda_2 (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}), \lambda_2 (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2}))$$

is a supersolution to Eq. [SI](#), satisfying $\bar{u} \geq \underline{u}$ and $\bar{v} \geq \underline{v}$. Indeed, as before, we will need two estimates. First, we estimate $\mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma)$. From Eq. [1.11](#), we obtain

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma) &\leq c_1 \lambda_2^{\frac{q_1}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma) + c_1 \lambda_2^{\frac{q_1}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma) \\ &\leq c_1 \lambda_2^{\frac{q_1}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma) + c_1 \lambda_2^{\frac{q_1}{p-1}} \lambda (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}), \end{aligned} \quad (2.11)$$

where $c_1 = c_1(p, q_1) > 0$. Notice that by Hölder's inequality and Young's inequality with the exponent γ_2 and its conjugate $\gamma_2' = ((p-1)(p-1+q_2))/(q_2(p-1+q_1))$, it follows

$$\begin{aligned} \int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma &\leq \left(\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma \right)^{\frac{1}{\gamma_2}} (\sigma(B(x,t)))^{\frac{1}{\gamma_2}} \\ &\leq c_2 \left(\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma + \sigma(B(x,t)) \right), \end{aligned}$$

where $c_2 = c_2(p, q_1, q_2) > 0$. This gives

$$\begin{aligned} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma)(x) &= \int_0^\infty \left(\frac{\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \\ &\leq c_2^{\frac{1}{p-1}} \int_0^\infty \left(\frac{(\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma + \sigma(B(x,t)))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \\ &\leq c_3 [\mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2 q_1} d\sigma)(x) + \mathbf{W}_{\alpha,p}\sigma(x)], \end{aligned}$$

where $c_3 = c_3(p, q_1, q_2) > 0$. Hence, by Eq. [1.11](#) again,

$$\mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_1} d\sigma)(x) \leq c_3 \mathbf{W}_{\alpha,p}\sigma(x) + c_3 \lambda (\mathbf{W}_{\alpha,p}\sigma(x) + (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_1}).$$

Applying the previous inequality in Eq. [2.11](#), we concluded that

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma) &\leq c_1 \lambda_2^{\frac{q_1}{p-1}} \left(c_3 \mathbf{W}_{\alpha,p}\sigma(x) + c_3 \lambda (\mathbf{W}_{\alpha,p}\sigma(x) + (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_1}) \right) \\ &\quad + c_1 \lambda_2^{\frac{q_1}{p-1}} \lambda (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}) \\ &\leq c_4 \lambda_2^{\frac{q_1}{p-1}} (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1}), \end{aligned} \quad (2.12)$$

where $c_4 = c_4(p, q_1, q_2, \lambda) > 0$. Next, similar arguments allow us to find an upper bound to $\mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma)$. Replacing q_1 by q_2 and γ_2 by γ_1 , from Eq. [1.11](#), we obtain

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma) &\leq \tilde{c}_1 \lambda_2^{\frac{q_2}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_2} d\sigma) + \tilde{c}_1 \lambda_2^{\frac{q_2}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1 q_2} d\sigma) \\ &\leq \tilde{c}_1 \lambda_2^{\frac{q_2}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_2} d\sigma) + \tilde{c}_1 \lambda_2^{\frac{q_2}{p-1}} \lambda (\mathbf{W}_{\alpha,p}\sigma + (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_2}), \end{aligned} \quad (2.13)$$

where $\tilde{c}_1 = \tilde{c}_1(p, q_2) > 0$. Also, we may deduce that

$$\begin{aligned} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{q_2} d\sigma)(x) &\leq \tilde{c}_2 \int_0^\infty \left(\frac{(\int_{B(x,t)} (\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1 q_2} d\sigma + \sigma(B(x,t)))}{t^{n-\alpha p}} \right)^{\frac{1}{p-1}} \frac{dt}{t} \\ &\leq \tilde{c}_3 [\mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p}\sigma)^{\gamma_1 q_2} d\sigma)(x) + \mathbf{W}_{\alpha,p}\sigma(x)], \\ &\leq \tilde{c}_3 [\lambda (\mathbf{W}_{\alpha,p}\sigma(x) + (\mathbf{W}_{\alpha,p}\sigma(x))^{\gamma_2}) + \mathbf{W}_{\alpha,p}\sigma(x)] \end{aligned}$$

where $\tilde{c}_2 > 0$ and $\tilde{c}_3 > 0$ depend only on p , q_1 and q_2 . Hence, from Eq. 2.13,

$$\begin{aligned} \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma) &\leq \tilde{c}_1 \lambda_2^{\frac{q_1}{p-1}} \left(\tilde{c}_3 \mathbf{W}_{\alpha,p} \sigma(x) + \tilde{c}_3 \lambda (\mathbf{W}_{\alpha,p} \sigma(x) + (\mathbf{W}_{\alpha,p} \sigma(x))^{\gamma_2}) \right) \\ &\quad + \tilde{c}_1 \lambda_2^{\frac{q_1}{p-1}} \lambda (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2}) \\ &\leq \tilde{c}_4 \lambda_2^{\frac{q_1}{p-1}} (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2}), \end{aligned} \quad (2.14)$$

where $\tilde{c}_4 = \tilde{c}_4(p, q_1, q_2, \lambda) > 0$. Therefore, choosing λ_2 such that

$$\lambda_2 = \max\{\lambda_1, c_4^{\frac{p-1}{p-1-q_1}}, (\tilde{c}_4)^{\frac{p-1}{p-1-q_2}}\},$$

we deduce from Eqs. 2.12 and 2.14 that

$$\begin{aligned} \bar{u} &\geq \underline{u}, & \bar{u} &\geq \mathbf{W}_{\alpha,p}(\bar{v}^{q_1} d\sigma), \\ \bar{v} &\geq \underline{v}, & \bar{v} &\geq \mathbf{W}_{\alpha,p}(\bar{u}^{q_2} d\sigma). \end{aligned}$$

This proves our claim.

The rest of the proof runs by using the same iterations argument given in Eq. 2.10 to guarantee the existence of a solution (u, v) to Eq. SI satisfying Eq. 1.10 for some $c = c(n, p, \alpha, q_1, q_2, \lambda) > 0$.

Now, to verify the reciprocal part of the theorem, let (u, v) be a nontrivial solution to Eq. SI satisfying Eq. 1.10 for some $c > 0$. By the lower bounds in Eq. 1.10,

$$\begin{aligned} u &= \mathbf{W}_{\alpha,p}(v^{q_1} d\sigma) \geq (c^{-1})^{\frac{q_1}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2 q_1} d\sigma), \\ v &= \mathbf{W}_{\alpha,p}(u^{q_2} d\sigma) \geq (c^{-1})^{\frac{q_2}{p-1}} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1 q_2} d\sigma). \end{aligned}$$

A combination of the previous estimates with the upper bounds in Eq. 1.10 yields

$$\begin{aligned} \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2 q_1} d\sigma) &\leq \lambda (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1}) < \infty \text{ - a.e.,} \\ \mathbf{W}_{\alpha,p}((\mathbf{W}_{\alpha,p} \sigma)^{\gamma_1 q_2} d\sigma) &\leq \lambda (\mathbf{W}_{\alpha,p} \sigma + (\mathbf{W}_{\alpha,p} \sigma)^{\gamma_2}) < \infty \text{ - a.e.,} \end{aligned}$$

where λ is given by $\lambda = \max\{c^{(p-1+q_1)/(p-1)}, c^{(p-1+q_2)/(p-1)}\}$. This completes the proof of Theorem 1.3.

3 k -Hessian equations

We begin this section by recalling some basic definitions and results due to N. S. Trudinger and X.-J. Wang [15–17] and D. A. Labutin [10]. These results were summarized in [13, Sect. 4] (see also [12, Sect. 7]), in which we state some of them for the convenience of the reader. Let Ω be an open set in $\mathbb{R}^n$. For $k = 1, \dots, n$ and $w \in C^2(\Omega)$, k -Hessian operator, $F_k[w]$ is defined as follows

$$F_k[w] = \sum_{1 \leq i_1 < \dots < i_k \leq n} \lambda_{i_1} \cdots \lambda_{i_k},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of the Hessian matrix $D^2 w$ on $\mathbb{R}^n$, with $k = 1, 2, \dots, n$. Alternatively, we may write $F_k[w] = [D^2 w]_k$, where $[A]_k$ denotes the sum of the $k \times k$ principal minors of an $n \times n$ matrix A . For example, $F_1[w] = \Delta w$ is the Laplacian operator of u , and $F_n[w] = \det D^2 w$ is the Monge-Ampère operator of w .

Definition 3.1 A function $w : \Omega \rightarrow (-\infty, \infty) \cup \{-\infty\}$ is k -subharmonic in Ω if

- (i) w is upper semicontinuous,
- (ii) $w \not\equiv -\infty$ in any component of Ω ,
- (iii) for each open subset D compactly contained in Ω and each $\varphi \in C^2(D) \cap C(\bar{D})$ satisfying $F_k[\varphi] \geq 0$ in D , the following implication holds

$$w \leq \varphi \text{ on } \partial D \implies w \leq \varphi \text{ in } D.$$

We denote by $\Phi^k(\Omega)$ the class of all k -subharmonic functions in Ω .

The following weak convergence result is essential to potential theory related to k -Hessian operators, and it was proved by N. S. Trudinger and X.-J. Wang [16, Theorem 1.1].

Theorem B For each $w \in \Phi^k(\Omega)$, there exists a Radon measure $\mu_k[w]$ such that

- (i) $\mu_k[w] = F_k[w]$ for $w \in C^2(\Omega)$, i.e. $\mu_k[w] = F_k[w] dx$, and
- (ii) if $\{w_j\}$ is a sequence of k -subharmonic functions which converges almost everywhere to w , then the sequence of the corresponding measures $\{\mu_k[w_j]\}$ converges to $\mu_k[w]$ weakly, that is,

$$\lim_{j \rightarrow \infty} \int_{\Omega} \varphi d\mu_k[w_j] = \int_{\Omega} \varphi d\mu_k[w] \quad \forall \varphi \in C_c^\infty(\Omega).$$

We mention that the measure $\mu_k[w]$ in Theorem B is said to be the k -Hessian measure associated with $w \in \Phi^k(\Omega)$, whence we may write $F_k[w]$ in place of $\mu_k[w]$ even in the case where w does not belong to $C^2(\Omega)$.

Remark 3.2 A Harnack-type convergence theorem follows from (ii) in Theorem B: If $\{w_j\} \subset \Phi^k(\Omega)$ is a nonincreasing sequence, then in each component of Ω the pointwise limit function $w = \lim_{j \rightarrow \infty} w_j$ is a k -subharmonic function and $\mu_k[w_j]$ converges to $\mu_k[w]$ weakly. Indeed, in each component of Ω , we have $w \in \Phi^k(\Omega)$ by definition of k -subharmonic functions.

Definition 3.3 For $\sigma \in M^+(\Omega)$, we say that w is a solution to the equation

$$F_k[w] = \sigma \quad \text{in } \Omega,$$

in the potential-theoretic sense, whenever

$$w \in \Phi^k(\Omega) \quad \text{and} \quad \mu_k[w] = \sigma.$$

The k -Hessian measure is an important tool in potential theory for $\Phi^k(\Omega)$. The next result is a direct consequence of [10, Theorem 2.1] due to D. A. Labutin, which brings a global pointwise estimate for functions in $\Phi^k(\Omega)$ in terms of Wolff potentials $\mathbf{W}_k$. See also [13, Theorem 4.3].

Theorem C Let $w \geq 0$ be such that $-w \in \Phi^k(\mathbb{R}^n)$, where $1 \leq k < n/2$. Assume that

$$\mu = F_k[-w] \quad \text{and} \quad \lim_{|x| \rightarrow \infty} w(x) = 0.$$

Then, there exists $K = K(n, k) > 0$ such that

$$K^{-1} \mathbf{W}_k \mu(x) \leq w(x) \leq K \mathbf{W}_k \mu(x) \quad \forall x \in \mathbb{R}^n.$$

In view of Definition 3.3, we present the precise definition of a solution to Syst. Eq. S in the potential-theoretic sense.

Definition 3.4 Let $1 \leq k \leq n$ and $q_1, q_2 > 0$. For $\sigma \in M^+(\mathbb{R}^n)$, we say that a pair of nonnegative function (u, v) is a solution to Syst. Eq. S whenever u and v satisfy

$$\begin{cases} -u \in \Phi^k(\mathbb{R}^n), & u \in L_{\text{loc}}^{q_2}(\Omega, d\sigma), \\ -v \in \Phi^k(\mathbb{R}^n), & v \in L_{\text{loc}}^{q_1}(\Omega, d\sigma), \\ d\mu_k[u] = v^{q_1} d\sigma, & d\mu_k[v] = u^{q_2} d\sigma, \\ \lim_{|x| \rightarrow \infty} u(x) = 0, & \lim_{|x| \rightarrow \infty} v(x) = 0. \end{cases}$$

This section will establish relevant existence results for Hessian equations with measure data. Following [14–16], $\Omega \subset \mathbb{R}^n$ is a *bounded uniformly $(k-1)$ -convex domain* in $\mathbb{R}^n$ whenever Ω is bounded, $\partial\Omega \in C^2$, and $H_j[\partial\Omega] > 0$ for $j = 1, \dots, k-1$, where $H_j[\partial\Omega]$ denotes the j -mean curvature of the boundary $\partial\Omega$. Note that a ball $B(x, R)$ is an example of bounded uniformly $(k-1)$ -convex domains in $\mathbb{R}^n$.

Let $\mu \in M^+(\Omega)$ and consider the following fully nonlinear problem:

$$\begin{cases} F_k[-w] = \mu & \text{in } \Omega, \\ w = 0 & \text{on } \partial\Omega, \end{cases} \quad (3.1)$$

where w is a nonnegative function such that $-w \in \Phi^k(\Omega)$ and it is continuous near $\partial\Omega$. Under a suitable assumption on μ , the existence of solutions to Eq. 3.1 is the content of the following lemma [13, Lemma 4.4].

Lemma D Let Ω be a bounded uniformly $(k-1)$ -convex domain in $\mathbb{R}^n$. Suppose $\mu \in M^+(\Omega)$ can be decomposed as a sum $\mu = f dx + \mu_0$, where μ_0 is a nonnegative measure compactly supported in Ω , and $f \geq 0$ belongs to $L^s(\Omega)$, for

- (i) $s > n/2$ if $1 \leq k \leq n/2$, or
- (ii) $s = 1$ if $n/2 < k \leq n$.

Then there exists $w \geq 0$ such that $-w \in \Phi^k(\Omega) \cap C(\overline{\Omega})$, and w solves Eq. 3.1.

In our argument to prove Theorem 1.5, we use a comparison principle due to N. S. Trudinger and X.-J. Wang in [17, Theorem 4.1]. For its statement, let us to introduce the so-called k -Hessian capacity $\text{cap}_k(\cdot, \Omega)$ defined for a compact set $E \subset \Omega$,

$$\text{cap}_k(E, \Omega) = \sup \left\{ \int_E d\mu_k[u] : u \in \Phi^k(\Omega), -1 < u < 0 \text{ in } \Omega \right\}.$$

We say that measure $\omega \in M^+(\Omega)$ is *continuous with respect to capacity* $\text{cap}_k(\cdot, \Omega)$ if $\omega(E) = 0$ whenever $\text{cap}_k(E, \Omega) = 0$.

Theorem E Let $w_1, w_2 \in \Phi^k(\Omega)$ such that $\mu_k[w_1]$ and $\mu_k[w_2]$ are continuous with respect to capacity $\text{cap}_k(\cdot, \Omega)$. Suppose $w_1 \leq w_2$ continuously on $\partial\Omega$. If $\mu_k[w_1] \geq \mu_k[w_2]$, then $w_1 \leq w_2$ in Ω .

Note that one can combine Lemma D with Theorem E to prove an existence and uniqueness result of solutions to Eq. 3.1, provided Ω is a bounded uniformly $(k-1)$ -convex domain in $\mathbb{R}^n$.

Remark 3.5 If $\sigma \in M^+(\mathbb{R}^n)$ satisfies the Riesz capacity condition Eq. 1.8 with $\alpha = 2k/(k+1)$ and $p = k+1$, then σ is continuous with respect to capacity $\text{cap}_k(\cdot) := \text{cap}_k(\cdot, \mathbb{R}^n)$.

Indeed, for $\alpha > 0$, we denote by $\mathbf{G}_\alpha \sigma$ the Bessel potential of order α , see [7, Section 1.2.4] for its definition. Similarly to Riesz capacity Eq. 1.6, we define the Bessel capacity of a compact set $E \subset \mathbb{R}^n$,

$$\text{cap}_{\mathbf{G}_\alpha, p}(E) = \inf\{\|f\|_{L^p}^p : f \in L^p(\mathbb{R}^n), f \geq 0, \mathbf{G}_\alpha f \geq 1 \text{ on } E\}.$$

By [12, Theorem 2.20], $\text{cap}_{\mathbf{G}_\alpha, p}$ is equivalent to the k -Hessian capacity cap_k for $\alpha = 2k/(k+1)$ and $p = k+1$, with $k < n/2$, i.e., there exists a constant $C = C(n, k) \geq 1$ such that, for all compact sets $E \subset \mathbb{R}^n$, it holds

$$C^{-1} \text{cap}_k(E) \leq \text{cap}_{\mathbf{G}_{\frac{2k}{k+1}}, k+1}(E) \leq C \text{cap}_k(E).$$

Since $\mathbf{G}_\alpha \sigma \leq \mathbf{I}_\alpha \sigma$ for all $\alpha > 0$ and $\sigma \in M^+(\mathbb{R}^n)$, one has $\text{cap}_{\mathbf{I}_\alpha, p} \leq \text{cap}_{\mathbf{G}_\alpha, p}$, whence the assertion follows.

The following result [13, Lemma 4.7] shows a monotonicity-type result to Hessian equations required in the argument in the next section. In what follows, we set $B_R = B(0, R)$ for $R > 0$.

Lemma F *Let $\mu_1, \mu_2 \in M^+(\mathbb{R}^n)$ satisfying $\mu_1 \leq \mu_2$, and $\mathbf{W}_k \mu_2 < \infty$. Let $w_1 \geq 0$ be a function satisfying $-w_1 \in \Phi^k(\mathbb{R}^n)$ and*

$$\begin{cases} F_k[-w_1] = \mu_1 & \text{in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} w_1 = 0. \end{cases}$$

Suppose w_1 is a limit of a subsequence of $\{w_1^i\}$ pointwise almost everywhere in $\mathbb{R}^n$, where w_1^i is continuous near ∂B_{i+1} and it satisfies

$$\begin{cases} -w_1^i \in \Phi^k(B_{i+1}), \\ F_k[-w_1^i] = \mu_{1,i} & \text{in } B_{i+1}, \\ w_1^i = 0 & \text{on } \partial B_{i+1}, \end{cases}$$

here $\mu_{1,i} := \chi_{B_i} \mu_1$. Then there exists a function w_2 satisfying

$$\begin{cases} w_2 \geq w_1, -w_2 \in \Phi^k(\mathbb{R}^n), \\ F_k[-w_2] = \mu_2 & \text{in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} w_2 = 0. \end{cases}$$

Moreover, w_2 is also a pointwise almost everywhere limit of a subsequence of $\{w_2^i\}$, where w_2^i is continuous near ∂B_{i+1} and it satisfies

$$\begin{cases} -w_2^i \in \Phi^k(B_{i+1}), \\ F_k[-w_2^i] = \mu_{2,i} & \text{in } B_{i+1}, \\ w_2^i = 0 & \text{on } \partial B_{i+1}. \end{cases}$$

4 Proofs of Theorem 1.5, Theorem 1.6, and Corollary 1.7

Before the proofs, we recall a basic fact on Wolff potentials (see [3, Corollary 3.2]): If $1 < p < \infty$ and $0 < \alpha < n/p$, for any $\omega \in M^+(\mathbb{R}^n)$, it holds

$$\lim_{|x| \rightarrow \infty} \mathbf{W}_{\alpha,p} \omega(x) = 0. \quad (4.1)$$

Thus, in view of Eq. 4.1, for any nontrivial solution (u, v) to Eq. 3.1 satisfying Eq. 1.10, one has

$$\lim_{|x| \rightarrow \infty} u(x) = \lim_{|x| \rightarrow \infty} v(x) = 0. \quad (4.2)$$

4.1 Proof of Theorem 1.5

Fix $1 \leq k < n/2$, and let K be the constant given in Theorem C. Let $\sigma \in M^+(\mathbb{R}^n)$ satisfying Eq. 1.8 with $\alpha = 2k/(k+1)$ and $p = k+1$. Applying Theorem 1.1 with $\alpha = 2k/(k+1)$, $p = k+1$ and $K^k \sigma$ in place of σ , there exists a nontrivial solution $(\tilde{u}, \tilde{v})$ to the system

$$\begin{cases} \tilde{u} = K \mathbf{W}_k(\tilde{v}^{q_1} d\sigma) & \text{in } \mathbb{R}^n, \\ \tilde{v} = K \mathbf{W}_k(\tilde{u}^{q_2} d\sigma) & \text{in } \mathbb{R}^n. \end{cases} \quad (4.3)$$

Moreover, $(\tilde{u}, \tilde{v})$ satisfies

$$\begin{aligned} c^{-1} (K \mathbf{W}_k \sigma)^{\gamma_1} &\leq \tilde{u} \leq c (K \mathbf{W}_k \sigma + (K \mathbf{W}_k \sigma)^{\gamma_1}), \\ c^{-1} (K \mathbf{W}_k \sigma)^{\gamma_2} &\leq \tilde{v} \leq c (K \mathbf{W}_k \sigma + (K \mathbf{W}_k \sigma)^{\gamma_2}), \end{aligned} \quad (4.4)$$

where γ_1 and γ_2 are given by Eq. 2.2, with $p-1 = k$, i.e.,

$$\gamma_1 = \frac{k(k+q_1)}{k^2 - q_1 q_2} \quad \text{and} \quad \gamma_2 = \frac{k(k+q_2)}{k^2 - q_1 q_2},$$

which can be rewritten as

$$\gamma_1 = \frac{q_1}{k} \gamma_2 + 1 \quad \text{and} \quad \gamma_2 = \frac{q_2}{k} \gamma_1 + 1.$$

Recall from Lemma 2.1 that $\mathbf{W}_k \sigma \in L^r_{\text{loc}}(\mathbb{R}^n, d\sigma)$ for all $r > 0$, whence $u_0, \tilde{u} \in L^{q_2}_{\text{loc}}(\mathbb{R}^n, d\sigma)$ and $v_0, \tilde{v} \in L^{q_1}_{\text{loc}}(\mathbb{R}^n, d\sigma)$.

We set $u_0 = \varepsilon (\mathbf{W}_k \sigma)^{\gamma_1}$ and $v_0 = \varepsilon (\mathbf{W}_k \sigma)^{\gamma_2}$, where $\varepsilon > 0$. Similarly to Claim 1, one can show $u_0 \leq K^{-1} \mathbf{W}_k(v_0^{q_1} d\sigma)$ and $v_0 \leq K^{-1} \mathbf{W}_k(u_0^{q_2} d\sigma)$ for

$$\varepsilon \leq \min\{(K^{-1} \kappa^{\frac{q_1 \gamma_2}{k}})^{\frac{k}{k-q_1}}, (K^{-1} \kappa^{\frac{q_2 \gamma_1}{k}})^{\frac{k}{k-q_2}}\},$$

where κ is the constant given in Lemma A. In addition, $u_0 \leq \tilde{u}$ and $v_0 \leq \tilde{v}$ if $\varepsilon \leq c^{-1} \min\{K^{\gamma_1}, K^{\gamma_2}\}$. $\varepsilon > 0$ is chosen to be sufficiently small throughout the proof.

Claim 2 There exist a pair of nonnegative functions $(u_1, v_1) \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$ satisfying

$$\begin{cases} -u_1, -v_1 \in \Phi^k(\mathbb{R}^n), \\ u_1 \geq u_0, \quad v_1 \geq v_0, \\ F_k[-u_1] = \sigma v_0^{q_1} \quad \text{in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} u_1(x) = 0, \\ F_k[-v_1] = \sigma u_0^{q_2} \quad \text{in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} v_1(x) = 0, \\ c_1(\mathbf{W}_k \sigma)^{\gamma_1} \leq u_1 \leq \tilde{u}, \quad \tilde{c}_1(\mathbf{W}_k \sigma)^{\gamma_2} \leq v_1 \leq \tilde{v} \quad \text{in } \mathbb{R}^n, \end{cases}$$

where

$$c_1 = K^{-1}(\varepsilon \kappa^{\gamma_2})^{\frac{q_1}{k}}, \quad \tilde{c}_1 = K^{-1}(\varepsilon \kappa^{\gamma_1})^{\frac{q_2}{k}}.$$

Indeed, letting $\sigma_{B_i} = \chi_{B_i} \sigma$, by Lemma D, there exist $u_1^i \geq 0$ and $v_1^i \geq 0$ satisfying

$$\begin{cases} -u_1^i \in \Phi^k(B_{i+1}) \cap C(\overline{B}_{i+1}), & -v_1^i \in \Phi^k(B_{i+1}) \cap C(\overline{B}_{i+1}), \\ F_k[-u_1^i] = \sigma_{B_i} v_0^{q_1} \quad \text{in } B_{i+1}, & F_k[-v_1^i] = \sigma_{B_i} u_0^{q_2} \quad \text{in } B_{i+1}, \\ u_1^i = 0 & \text{on } \partial B_{i+1}, & v_1^i = 0 & \text{on } \partial B_{i+1}. \end{cases}$$

Using [13, Theorem 4.2], we obtain

$$\begin{aligned} u_1^i &\leq K \mathbf{W}_k(v_0^{q_1} d\sigma) \leq K \mathbf{W}_k(\tilde{v}^{q_1} d\sigma) = \tilde{u}, \\ v_1^i &\leq K \mathbf{W}_k(u_0^{q_2} d\sigma) \leq K \mathbf{W}_k(\tilde{u}^{q_2} d\sigma) = \tilde{v} \quad \forall i = 1, 2, \dots \end{aligned} \quad (4.5)$$

Extending by zero away from B_{i+1} , we may actually assume that u_1^i and v_1^i are functions in whole $\mathbb{R}^n$. It follows from Theorem E and Remark 3.5 that the sequences $\{u_1^i\}$ and $\{v_1^i\}$ are increasing pointwise in $\mathbb{R}^n$. Consequently, by Remark 3.2, the pointwise limits $u_1 := \lim_{i \rightarrow \infty} u_1^i$ and $v_1 := \lim_{i \rightarrow \infty} v_1^i$ belong to $\Phi^k(\mathbb{R}^n)$, with $\{F_k[-u_1^i]\}_i$ and $\{F_k[-v_1^i]\}_i$ converging weakly to $F_k[-u_1]$ and $F_k[-v_1]$, respectively. On the other hand, by Monotone Convergence Theorem, $\{\sigma_{B_i} v_0^{q_1}\}_i$ and $\{\sigma_{B_i} u_0^{q_2}\}_i$ converge weakly to $\sigma v_0^{q_1}$ and $\sigma u_0^{q_2}$, respectively. Hence, as Radon measures in $\mathbb{R}^n$,

$$F_k[-u_1] = \sigma v_0^{q_1} \quad \text{and} \quad F_k[-v_1] = \sigma u_0^{q_2}.$$

Next, from Eq. 4.5, one has $u_1 \leq \tilde{u}$ and $v_1 \leq \tilde{v}$, whence $u_1 \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma)$, $v_1 \in L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$, and by Eq. 4.2

$$\lim_{|x| \rightarrow \infty} u_1(x) = \lim_{|x| \rightarrow \infty} v_1(x) = 0.$$

Combining Theorem C with Lemma A, we deduce

$$\begin{aligned} u_1 &\geq K^{-1} \mathbf{W}_k(v_0^{q_1} d\sigma) = K^{-1} \varepsilon^{\frac{q_1}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_2 q_1} d\sigma) \\ &\geq K^{-1} \varepsilon^{\frac{q_1}{k}} \kappa^{\frac{q_1}{k} \gamma_2} (\mathbf{W}_k \sigma)^{\frac{q_1}{k} \gamma_2 + 1} = K^{-1} \varepsilon^{\frac{q_1}{k}} \kappa^{\frac{q_1}{k} \gamma_2} (\mathbf{W}_k \sigma)^{\gamma_1}, \\ v_1 &\geq K^{-1} \mathbf{W}_k(u_0^{q_2} d\sigma) = K^{-1} \varepsilon^{\frac{q_2}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_1 q_2} d\sigma) \\ &\geq K^{-1} \varepsilon^{\frac{q_2}{k}} \kappa^{\frac{q_2}{k} \gamma_1} (\mathbf{W}_k \sigma)^{\frac{q_2}{k} \gamma_1 + 1} = K^{-1} \varepsilon^{\frac{q_2}{k}} \kappa^{\frac{q_2}{k} \gamma_1} (\mathbf{W}_k \sigma)^{\gamma_2}, \end{aligned}$$

By choice of $\varepsilon > 0$, the previous inequalities give $u_1 \geq u_0$ and $v_1 \geq v_0$. This completes the proof Claim 2.

With the aid of Lemma F, analysis similar to that in the proof of Claim 2 allows us to construct, inductively, a sequence of pairs of nonnegative functions $\{(u_j, v_j)\} \subset L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma) \times L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$ such that

$$\begin{cases} -u_j, -v_j \in \Phi^k(\mathbb{R}^n), \\ F_k[-u_j] = \sigma v_{j-1}^{q_1} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} u_j(x) = 0, \\ F_k[-v_j] = \sigma u_{j-1}^{q_2} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} v_j(x) = 0, \\ 0 \leq u_j \leq u_{j+1}, \quad 0 \leq v_j \leq v_{j+1} \text{ in } \mathbb{R}^n \\ c_j (\mathbf{W}_k \sigma)^{\gamma_1} \leq u_j \leq \tilde{u}, \quad \tilde{c}_j (\mathbf{W}_k \sigma)^{\gamma_2} \leq v_j \leq \tilde{v} \text{ in } \mathbb{R}^n. \end{cases} \quad (4.6)$$

where c_1 and $\tilde{c}_1$ are as above, and for $j = 2, 3, \dots$

$$c_j = K^{-1}(\tilde{c}_{j-1} \kappa^{\gamma_2})^{\frac{q_1}{k}}, \quad \tilde{c}_j = K^{-1}(c_{j-1} \kappa^{\gamma_1})^{\frac{q_2}{k}}. \quad (4.7)$$

Indeed, let u_1 and v_1 as in Claim 2. Since $\sigma v_1^{q_1} \geq \sigma v_0^{q_1}$ and $\sigma u_1^{q_2} \geq \sigma u_0^{q_2}$, it follows from Lemma F that there exist a pair of nonnegative functions (u_2, v_2) satisfying

$$\begin{cases} -u_2, -v_2 \in \Phi^k(\mathbb{R}^n) \\ u_2 \geq u_1, \quad v_2 \geq v_1, \\ F_k[-u_2] = \sigma v_1^{q_1} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} u_2(x) = 0, \\ F_k[-v_2] = \sigma u_1^{q_2} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} v_2(x) = 0. \end{cases}$$

Applying Theorem C, one has

$$\begin{aligned} u_2 &\leq K \mathbf{W}_k(v_1^{q_1} d\sigma) \leq K \mathbf{W}_k(\tilde{v}^{q_1} d\sigma) = \tilde{u}, \\ v_2 &\leq K \mathbf{W}_k(u_1^{q_2} d\sigma) \leq K \mathbf{W}_k(\tilde{u}^{q_2} d\sigma) = \tilde{v}, \end{aligned}$$

and, by Lemma A,

$$\begin{aligned} u_2 &\geq K^{-1} \mathbf{W}_k(v_1^{q_1} d\sigma) = K^{-1} \tilde{c}_1^{\frac{q_1}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_2 q_1} d\sigma) \\ &\geq K^{-1} \tilde{c}_1^{\frac{q_1}{k}} \kappa^{\frac{q_1}{k} \gamma_2} (\mathbf{W}_k \sigma)^{\frac{q_1}{k} \gamma_2 + 1} = c_2 (\mathbf{W}_k \sigma)^{\gamma_1}, \\ v_2 &\geq K^{-1} \mathbf{W}_k(u_1^{q_2} d\sigma) = K^{-1} c_1^{\frac{q_2}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_1 q_2} d\sigma) \\ &\geq K^{-1} c_1^{\frac{q_2}{k}} \kappa^{\frac{q_2}{k} \gamma_1} (\mathbf{W}_k \sigma)^{\frac{q_2}{k} \gamma_1 + 1} = \tilde{c}_2 (\mathbf{W}_k \sigma)^{\gamma_2}. \end{aligned}$$

Now, suppose (u_j, v_j) has been obtained for some $j > 1$. In the same manner, one shows that there exist a pair of nonnegative functions (u_{j+1}, v_{j+1}) satisfying

$$\begin{cases} -u_{j+1}, -v_{j+1} \in \Phi^k(\mathbb{R}^n) \\ u_{j+1} \geq u_j, \quad v_{j+1} \geq v_j, \\ F_k[-u_{j+1}] = \sigma v_j^{q_1} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} u_{j+1}(x) = 0, \\ F_k[-v_{j+1}] = \sigma u_j^{q_2} \text{ in } \mathbb{R}^n, \quad \lim_{|x| \rightarrow \infty} v_{j+1}(x) = 0. \\ c_{j+1} (\mathbf{W}_k \sigma)^{\gamma_1} \leq u_{j+1} \leq \tilde{u}, \quad \tilde{c}_{j+1} (\mathbf{W}_k \sigma)^{\gamma_2} \leq v_{j+1} \leq \tilde{v}. \end{cases}$$

From the last line in Eq. 4.6, we deduce that $(u_j, v_j) \in L_{\text{loc}}^{q_2}(\Omega, d\sigma) \times L_{\text{loc}}^{q_1}(\Omega, d\sigma)$ for all $j \geq 1$.

We set the pointwise limits $u = \lim_{j \rightarrow \infty} u_j$ and $v = \lim_{j \rightarrow \infty} v_j$. Using Remark 3.2 again, both $-u$ and $-v$ belong to $\Phi^k(\mathbb{R}^n)$. By Monotone Convergence Theorem, $u \in L_{\text{loc}}^{q_2}(\mathbb{R}^n, d\sigma)$ and $v \in L_{\text{loc}}^{q_1}(\mathbb{R}^n, d\sigma)$. Thus (u, v) satisfies

$$\begin{cases} F_k[-u] = \sigma v^{q_1} & \text{in } \mathbb{R}^n, \\ F_k[-v] = \sigma u^{q_2} & \text{in } \mathbb{R}^n. \end{cases}$$

In view of the last line in Eq. 4.6, letting $j \rightarrow \infty$, one can see that $u \leq \tilde{u}$ and $v \leq \tilde{v}$, whence $\lim_{|x| \rightarrow \infty} u(x) = \lim_{|x| \rightarrow \infty} v(x) = 0$, which shows that (u, v) solves Syst. Eq. S in sense of Definition 2.9. Furthermore, there exists a constant $C_1 = C_1(n, k, q_1, q_2, C_\sigma) > 0$ such that

$$u \leq C_1 (\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\gamma_1}) \quad \text{and} \quad v \leq C_1 (\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\gamma_2}),$$

which assures the upper bound in Eq. 1.12. Since $u \geq c_j (\mathbf{W}_k \sigma)^{\gamma_1}$ and $v \geq \tilde{c}_j (\mathbf{W}_k \sigma)^{\gamma_2}$ for all $j \geq 1$, a passage to the limit $j \rightarrow \infty$ implies that

$$u \geq C_2 (\mathbf{W}_k \sigma)^{\gamma_1} \quad \text{and} \quad v \geq \tilde{C}_2 (\mathbf{W}_k \sigma)^{\gamma_2},$$

where is straightforward to compute

$$\begin{aligned} C_2 &= \lim_{j \rightarrow \infty} c_j = K^{-\gamma_1} \kappa^{\frac{\gamma_1 q_1 (\gamma_1 q_2 + \gamma_2 k)}{k^2}}, \\ \tilde{C}_2 &= \lim_{j \rightarrow \infty} \tilde{c}_j = K^{-\gamma_2} \kappa^{\frac{\gamma_2 q_2 (\gamma_2 q_1 + \gamma_1 k)}{k^2}}, \end{aligned}$$

which shows a lower bound in Eq. 1.12. This completes the proof of Theorem 1.5.

4.2 Proof of Theorem 1.6

Fix $1 \leq k < n/2$. Let γ_1 and γ_2 given by Eq. 2.2. We begin by proving the statement (i). Let (u, v) be a nontrivial solution to Eq. S, in the sense of Definition 2.9, satisfying Eq. 1.12. By Theorem C, (u, v) satisfies the system

$$\begin{cases} u \geq K^{-1} \mathbf{W}_k(v^{q_1} d\sigma), \\ v \geq K^{-1} \mathbf{W}_k(u^{q_2} d\sigma). \end{cases}$$

Applying Eq. 1.12 in the previous inequalities,

$$\begin{aligned} K^{-1} c^{-\frac{q_1}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_2 q_1} d\sigma) &\leq K^{-1} \mathbf{W}_k(v^{q_1} d\sigma) \leq u \leq c (\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\gamma_1}), \\ K^{-1} c^{-\frac{q_2}{k}} \mathbf{W}_k((\mathbf{W}_k \sigma)^{\gamma_1 q_2} d\sigma) &\leq K^{-1} \mathbf{W}_k(u^{q_2} d\sigma) \leq v \leq c (\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\gamma_2}). \end{aligned}$$

Letting $\lambda = K \max\{c^{\frac{k+q_1}{k}}, c^{\frac{k+q_2}{k}}\}$, the statement (i) follows.

To establish the statement (ii), let $\lambda > 0$ such that Eq. 1.13 holds. By Theorem 1.3 with $\alpha = 2k/(k+1)$, $p = k+1$ and $K^k \sigma$ in place of σ , there exists a nontrivial solution $(\tilde{u}, \tilde{v})$ to Syst. Eq. 4.3, satisfying Eq. 4.4 with $c = c_0(n, k, q_1, q_2, \lambda)$. As in the proof of Theorem 1.5, the existence of such $(\tilde{u}, \tilde{v})$ yields that there exists a solution (u, v) to Eq. S for which Eq. 1.12 holds with $c = c_1(n, k, q_1, q_2, \lambda) > 0$. This proves the statement (ii) and completes the proof of Theorem 1.6.

4.3 Proof of Corollary 1.7

Fix $1 \leq k < 2$, and let $\sigma \in M^+(\mathbb{R}^n)$. Suppose $q_1 = q_2 < k$. Setting $q = q_1$, from Eq. 2.2 γ_1 and γ_2 satisfy

$$\gamma_1 = \gamma_2 = \frac{k(k+q)}{k^2 - q^2} = \frac{k}{k-q}. \quad (4.8)$$

The statements (i), (ii) and (iii) follow by the same methods as in the proofs of Theorem 1.5 and Theorem 1.6, respectively.

Indeed, suppose Eq. 1.8. Following the notation used in the proof of Theorem 1.5, Eq. 4.8 yields $u_0 = v_0$, whence $u_1 = v_1$ by Claim 2. Consequently, by induction

$$u_j = v_j \quad \forall j = 1, 2, \dots,$$

where (u_j, v_j) are given in Eq. 4.6 for $j = 1, 2, \dots$. A passage to the limit as before implies that $u = v \in L_{\text{loc}}^q(\mathbb{R}^n, d\sigma)$ and $-u \in \Phi^k(\mathbb{R}^n)$. Hence Syst. Eq. S gives Eq. P, i.e.

$$\begin{cases} F_k[-u] = \sigma u^q & \text{in } \mathbb{R}^n, \\ \lim_{|x| \rightarrow \infty} u(x) = 0. \end{cases}$$

Furthermore, u satisfies bilateral estimates Eq. 1.14, with $c = c(n, k, q, \sigma) > 0$. This shows part (i). The statement (ii) follows from combining the lower bound given in Theorem C with Eq. 1.14,

$$K^{-1} c^{-\frac{q}{k}} \mathbf{W}_k \left((\mathbf{W}_k \sigma)^{\frac{kq}{k-q}} d\sigma \right) \leq K^{-1} \mathbf{W}_k(u^q d\sigma) \leq u \leq c \left(\mathbf{W}_k \sigma + (\mathbf{W}_k \sigma)^{\frac{k}{k-q}} \right),$$

whence we may take $\lambda = K c^{\frac{k+q}{k}}$. This establishes the assertion. To prove the statement (iii), we first notice that a combination of Remark 1.4 with Eq. 1.15 implies that there exists a solution $\tilde{u} \in L_{\text{loc}}^q(\mathbb{R}^n, d\sigma)$ to equation

$$\tilde{u} = K \mathbf{W}_k(\tilde{u}^q d\sigma) \quad \text{in } \mathbb{R}^n.$$

Thus, as in the proof of Theorem 1.6, there exists a solution u to Eq. P satisfying Eq. 1.14, with $c = c(n, k, q, \lambda) > 0$. This verify (iii), and finally completes the proof of Corollary 1.7.

Acknowledgements Research supported in part by CNPq and CAPES/Brazil.

Author Contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by all authors. The authors read and approved the final manuscript.

Funding E. da Silva acknowledges partial support from CNPq through grants 140394/2019-2 and J.M. do Ó acknowledges partial support from CNPq through grants 312340/2021-4 and 429285/2016-7 and Paraíba State Research Foundation (FAPESQ), grant no 3034/2021.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval All data generated or analyzed during this study are included in this article.

Consent to participate All authors consent to participate in this work.

Conflict of interest The authors declare no conflict of interest.

Consent for publication All authors consent for publication.

Competing interests The authors declare no competing interests.

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